

EmPay – Project File Structure

Monorepo layout with owner annotations for all 4 team members

Colour Legend (row background = file owner):

Mohak

Frontend + Testing

Manav

Full Stack Bridge

Vismay

Frontend UI

Kishan

Backend + DevOps

[+] = Folder - = File [DIR] prefix = directory Row colour = primary owner Note column = what the file does

Path / File	Purpose & Notes
[+] empay/	Monorepo root
[+] client/	React + Vite frontend
[+] public/	Static assets, favicon
[+] src/	All source code
[+] api/	Axios call wrappers — Manav owns all
- authApi.js	login(), register(), getMe()
- attendanceApi.js	markAttendance(), fetchLogs(), fetchStats()
- leaveApi.js	apply(), approve(), reject(), getBalance()
- payrollApi.js	triggerPayrun(), getPayslip(), downloadPdf()
- employeeApi.js	getAll(), create(), update(), delete()
- aiApi.js	suggest() — calls /api/ai/suggest (Haiku)
[+] components/	
[+] ui/	shadcn primitives: Button, Card, Modal, Badge
[+] layout/	Shared layout components — Mohak
- Sidebar.jsx	Role-aware nav links
- Navbar.jsx	Top bar with user info + logout
- ProtectedRoute.jsx	Redirects if no token / wrong role
[+] attendance/	Attendance components — Mohak
- MarkAttendance.jsx	Check-in / check-out button + live timer
- AttendanceLogs.jsx	Calendar heatmap of daily/monthly logs
[+] leave/	Leave components — Mohak + Vismay
- LeaveApplyForm.jsx	Type selector, date picker, AI reason chip
- LeaveStatus.jsx	Employee: own leave requests + badges
- LeaveApprovalTable.jsx	Payroll Officer: approve / reject table
[+] payroll/	Payroll components — Manav + Vismay
- PayrunList.jsx	Table of payruns, trigger new payrun button

- PayslipView.jsx	Formatted payslip with all salary fields
- SalaryBreakdown.jsx	Basic / HRA / Special / PF / Net display
[+] dashboard/	Role-specific dashboards — Vismay
- AdminDashboard.jsx	Stat cards + bar chart + AI insight card
- HRDashboard.jsx	Attendance overview table, leave count
- PayrollDashboard.jsx	Payrun status, total payout, payslip button
- EmployeeDashboard.jsx	Own attendance %, leave balance, payslip
[+] employees/	Employee management — Vismay
- EmployeeList.jsx	Searchable, filterable employee table
- EmployeeForm.jsx	Create / edit employee (Admin + HR only)
- EmployeeProfile.jsx	Read-only profile card for all roles
[+] pages/	Route-level page components
- Login.jsx	Email + password form, redirect on success
- Register.jsx	Admin-facing user creation page
- Dashboard.jsx	Reads role from store, renders correct dash
- AttendancePage.jsx	Wraps MarkAttendance + AttendanceLogs
- LeavePage.jsx	Wraps LeaveApplyForm + LeaveStatus
- PayrollPage.jsx	Wraps PayrunList + PayslipView
- EmployeesPage.jsx	Wraps EmployeeList + EmployeeForm
- ProfilePage.jsx	View + edit own profile, avatar
[+] store/	Zustand global state — Manav
- authStore.js	user, role, token, login(), logout()
- notifStore.js	Toast queue for success/error messages
[+] hooks/	React Query custom hooks — Manav
- useAttendance.js	useQuery + useMutation for attendance
- useLeave.js	useQuery + useMutation for leaves
- usePayroll.js	useQuery + useMutation for payroll
[+] utils/	Pure helper functions — Mohak
- payrollCalc.js	CLIENT-SIDE display only: format net pay
- dateHelpers.js	formatDate(), getWorkingDays(), monthRange()
- roleGuard.js	Map of role -> allowed route paths
- App.jsx	Router setup, QueryClientProvider, Zustand
- main.jsx	React DOM entry point
- .env	VITE_API_URL, VITE_AI_ENABLED
- package.json	deps: react, tailwind, shadcn, recharts, zod
- vite.config.js	proxy /api -> :5000, /ai -> :5000
[+] server/	Node.js + Express backend
[+] config/	
- db.js	Mongoose.connect() with retry logic
- env.js	dotenv loader + validation
[+] models/	Mongoose schemas — Kishan
- User.js	name, email, passwordHash, role, employeeId

- Employee.js	dept, designation, wage, joinDate, pfOptIn
- Attendance.js	employeeId, date, checkIn, checkOut, status
- Leave.js	empId, type, startDate, endDate, status, approvedBy
- LeaveBalance.js	empId, annual, sick, casual (days remaining)
- Payrun.js	month, year, status, createdBy, totalAmount
- Payslip.js	payrunId, empId, earnings{}, deductions{}, net
[+] routes/	Express routers — Kishan
- auth.routes.js	POST /register /login GET /me
- employee.routes.js	CRUD /employees with role middleware
- attendance.routes.js	POST /mark GET /logs GET /stats
- leave.routes.js	apply, approve, reject, list, balance
- payroll.routes.js	payrun CRUD, payslip get, pdf download
- ai.routes.js	POST /suggest — Haiku proxy — Manav
[+] controllers/	Business logic — Kishan + Manav
- auth.controller.js	register: hash pw + create User; login: jwt
- employee.controller.js	CRUD with role checks
- attendance.controller.js	mark: upsert per day; logs: filter by role
- leave.controller.js	apply: deduct balance; approve/reject: notify
- payroll.controller.js	calls payrollEngine.js for each employee
- ai.controller.js	POST to Anthropic API, return suggestion
[+] middleware/	
- auth.middleware.js	verifyToken: decode JWT, attach req.user
- role.middleware.js	requireRole('admin','hr') factory function
[+] utils/	
- payrollEngine.js	THE source of truth: PF, profTax, pro-rated
- server.js	Express app init, route mounting, listen
- .env	MONGO_URI, JWT_SECRET, ANTHROPIC_API_KEY
- package.json	deps: express, mongoose, bcrypt, jsonwebtoken
[+] pdf-service/	Python FastAPI microservice — Kishan
- main.py	FastAPI app, POST /generate-payslip endpoint
[+] templates/	
- payslip_template.py	ReportLab renderer: earnings/deductions/net
- requirements.txt	fastapi, uvicorn, reportlab
- .env	PORT=8001
[+] docs/	Documentation — Mohak writes after testing
- SETUP.md	How to run client, server, pdf-service
- API.md	All endpoints, params, responses
- PAYROLL_LOGIC.md	Formula documentation with examples

Service Port Reference

Service	Port	Command	Owner
React (Vite)	:3000	npm run dev	Mohak / Vismay

Service	Port	Command	Owner
Node/Express	:5000	npm run dev (nodemon)	Kishan / Manav
Python PDF service	:8001	uvicorn main:app --port 8001	Kishan
MongoDB Atlas	cloud	Connection via MONGO_URI in .env	Kishan

.env Variables Reference

Variable	File	Value / Description
VITE_API_URL	client/.env	http://localhost:5000
VITE_AI_ENABLED	client/.env	true (toggle Haiku features)
MONGO_URI	server/.env	MongoDB Atlas connection string
JWT_SECRET	server/.env	Any strong random string (32+ chars)
ANTHROPIC_API_KEY	server/.env	From GitHub Education Pack / Anthropic Console
PORT	pdf-service/.env	8001